

Isha Jain

Junior Undergraduate | Department of Electrical Engineering
Incoming SDE intern, Google

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EDUCATION

Year	Degree	Institute	CPI/%
2021 - Present	B.Tech	Indian Institute of Technology Kanpur	7.3/10
2021	Class XII (CBSE)	Saarthi International School	94.6
2019	Class X (CBSE)	Delhi Public School	98

ACHIEVEMENTS

- Selected in **top 50** teams of Tata Steel-Women of Mettle hackathon which provided an opportunity to work on real-time project
- Received the **Best Incoming Sportsperson** (Aquatics) Award 2022-2023 from Games and Sports Council, IIT Kanpur
- Won a **Silver medal** in Swimming and contributed to the overall first runner-up (women) in **Inter IIT Aquatics Meet 2022**
- Secured an **All India Rank 2578** in Joint Entrance Examination - Advanced 2021 among 0.2 million students
- Secured an **All India Rank 5782** in Joint Entrance Examination - Mains 2021 among 1.2 million students
- Qualified to the second stage of National Talent Search Examination (**NTSE**) among top **100** in Uttar Pradesh in 2019
- Won a **Bronze medal** in Swimming in CBSE Nationals (North zone) among **250+** participants in October'15

WORK EXPERIENCE

- MICROSCOPIC IMAGE ANALYSIS** | *SURGE'23* | *Mentor: Prof. Tushar Sandhan, IIT Kanpur* | 🌐 *May'23 - Present*

Objective	• Analyze high resolution cultured mammalian cells image and detect mitochondrial networks in the dataset
Approach	• Utilized ImageJ software for preprocessing, including median filtering , CLAHE , & unsharp masking • Applied Hough Circle Transform which involves edge detection, circle accumulator, voting and thresholding • Utilized data augmentation and Labkit plugin, implemented pre-trained model for accurate edge detection
Outcome	• Detected circular mitochondria and currently working on developing an algorithm to detect convex polygon

KEY PROJECTS

- GAIT ANALYSIS** | *Mentor: Prof. Tushar Sandhan, Department of Electrical Engineering, IIT Kanpur* | 🌐 *Dec'22 - Jun'23*
 - Developed **Image Processing** algorithm to analyze gait, used **OpenCV** library to crop dataset images into bounding box
 - Extracted features using time series and implemented **ANN** model to accurately classify the dataset of **20** individuals
 - Analyzed cropped images using their **optical flow** and computed merged HSV optical flow with the cropped average image
 - Implemented pre-trained **VGG19** (CNN network) of **Tensorflow** library to achieve the classification accuracy of **98.75%**
- BUSINESS ANALYTICS USING PYTHON** | *Finance and Analytics Club, IIT Kanpur* | 🌐 *May'22 - Aug'22*
 - Implemented and processed historical business data, analyzed that data to identify trends, patterns and root causes
 - Successfully implemented various **python** libraries, learned concepts like **regression**, **Decision tree** and cluster Analysis
 - Developed **music recommendation system** by analyzing user's listened songs, suggesting similar songs based on data points
- EAVESDROPPING** | *Stamatics, IIT Kanpur* | 🌐 *May'23 - Jul'23*
 - Successfully analyzed **vigenere cipher** substitution and **permutation cipher** techniques for encryption and decryption
 - Developed in-depth understanding of cryptanalysis techniques for **DES**, **AES**, and **RSA** encryption algorithms

TECHNICAL SKILLS

- Programming Skills:** C, C++, Python, HTML, CSS
- Frameworks & Libraries:** NumPy, Pandas, Matplotlib, Tensorflow, OpenCV, Scikit-learn
- Softwares & Utilities:** Git, L^AT_EX, Fusion 360, ImageJ, Canva, MS Office, VS code

RELEVANT COURSES

Fundamentals of Computing Complex Variables Microelectronics I	Data Structure & Algorithms* Probability and Statistics Introduction to Electronics	Linear Algebra & Differential Equations Control System Analysis Signals, Systems and Networks
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POSITIONS OF RESPONSIBILITY

- Captain, Aquatics Team** | *Games and Sports Council, IIT Kanpur* *Apr'23 - Present*

Leadership	• Led and managed institute Aquatics team , fostering unified environment for skill development and competition • Performed regular assessments and evaluations to ensure the availability of adequate equipment for the team
Initiatives	• Conducted trials with coach, evaluating the skills and abilities of incoming players for selection into the team • Organized and led 60 days long summer camp mentoring and training 20+ athletes from different batches • Improved the team bonding and ensured team fitness by conducting team meetings and batch farewell
Impact	• The average attendance of the students increased from 60% to 80% throughout the summer camp • Players improved timing and successfully learned to swim proficiently in multiple strokes in summer camp
- Secretary, Events** | *Entrepreneurship Cell, IIT Kanpur* *May'22 - Apr'23*
 - Conducted two campus Hangouts with the alumni to get insights from their startup journey with an audience of **150+**
 - Worked as a part of the 3-tier team managing and organizing various events in E-summit 2022 with **15000+** participants
- Secretary, Outreach and Databasing** | *Electrical Engineering Association, IIT Kanpur* *Nov'22 - Present*
 - Organised sessions for junior batch giving them brief about Electrical Engineering in IIT Kanpur and career opportunities
 - Maintained comprehensive **database** of alumni from the Electrical Department and organized **farewell** for graduating seniors

EXTRA-CURRICULAR ACTIVITIES

- Mentored 6 freshmen, facilitating their personal and academic journey for a smooth college transition as a **Student Guide**
- Represented Hall-6 in the **Inter-Hall** technical competition, Takneek and tackled a **web development** problem statement